

KELT-23Ab

R-2221

Benchmark run · workflow W8-benchmark-deep · record deep-bench_KELT-23A_250611 · created 2026-08-14T23:45:04+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460837.71158 +/- 0.00088 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.105 +/- 0.012
Transit depth (Rp/Rs)^2	1.1 +/- 0.24 [%]
Orbital Inclination (inc)	83.55 +/- 0.36
Airmass coefficient 1 (a1)	0.867 +/- 0.0043
Airmass coefficient 2 (a2)	0.115 +/- 0.054
Scatter in the residuals of the lightcurve fit is	0.49 %
Best Comparison Star	None
Optimal Aperture	4.38
Optimal Annulus	16.42
Transit Duration (day)	0.0638 +/- 0.0041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.105 ± 0.012 vs 0.132 (deviation 1.07σ)
Duration fit vs published	0.0638 d vs 0.0954 d (deviation 3.68σ)
Mid-time O–C	0.9 min
Depth SNR	4.2
Residual scatter	0.49 %

Light curve

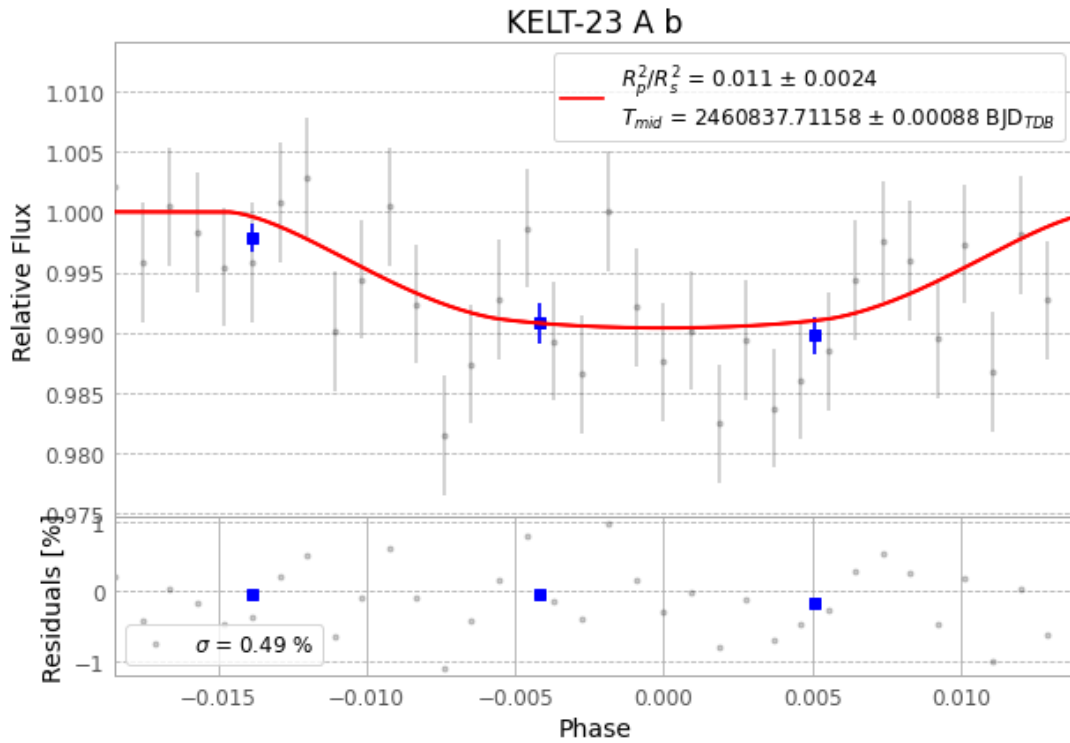


Figure 1 — FinalLightCurve_KELT-23 A b_2025-06-10.png (SHA-256 b7f64f778b02ddfc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [388, 238], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 96da580d8a2d8247...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

36 FITS frames (24 MB), KELT-23A250611040316.FITS → KELT-23A250611054815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KELT-23A-250611.log (40ffcd232e2b...), FinalLightCurve_KELT-23 A b_2025-06-10.png (b7f64f778b02...), FinalLightCurve_KELT-23 A b_2025-06-10.csv (ae2ecbb8d83d...)

Compute resources

Wall time 133.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 23:45Z.